SLR disk enrichment through AGB interloper stars

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Context

- **Short-Lived Radioisotopes (SLRs; such as ^{26}Al and ^{60}Fe) drive out-gassing and devolatilization in planetesimals.**
- $\sim$1/10th of initial Solar ^{26}Al abundance can strip volatiles from planetesimals (Eatson et al., 2024a; Lichtenberg et al., 2019).
- The source of these SLRs is not well established, there are issues with each theory:
  - Cosmic ray spallation causes inhomogeneous SLR distribution and insufficient yields.
  - Massive stars do not produce enough ^{60}Fe (Eatson et al., 2024b), and their UV flux disrupts protoplanetary disks (Haworth et al., 2023).
  - Supernovae occur after the bulk of planetesimal production (Eatson et al., 2024b).
- **Parker and Schoettler (2023) found an example of an AGB star "interloping" with a cluster, which could produce Solar-like SLR enrichment in low-mass cluster disks without hypersonic winds or destructive levels of UV flux.**

Methodology

- **We performed a rigorous parameter space search to determine the conditions necessary for efficient SLR enrichment.**
- AMUSE (Pelupessy et al., 2013) was used to produce a series of 100-star clusters.
- A $7\,\mathrm{M}_\odot$ AGB was then added to each simulation, travelling in the direction of the cluster.
- SLR outflow was modelled by a series of cylindrical segments from the interloper's path.
- Disks that intersected cylinder segments sweep up available SLRs.
- Simulations were repeated numerous times in order to build up a statistical population of low-mass clusters.
- AGB SLR yields were derived from Karakas and Lugaro (2016).

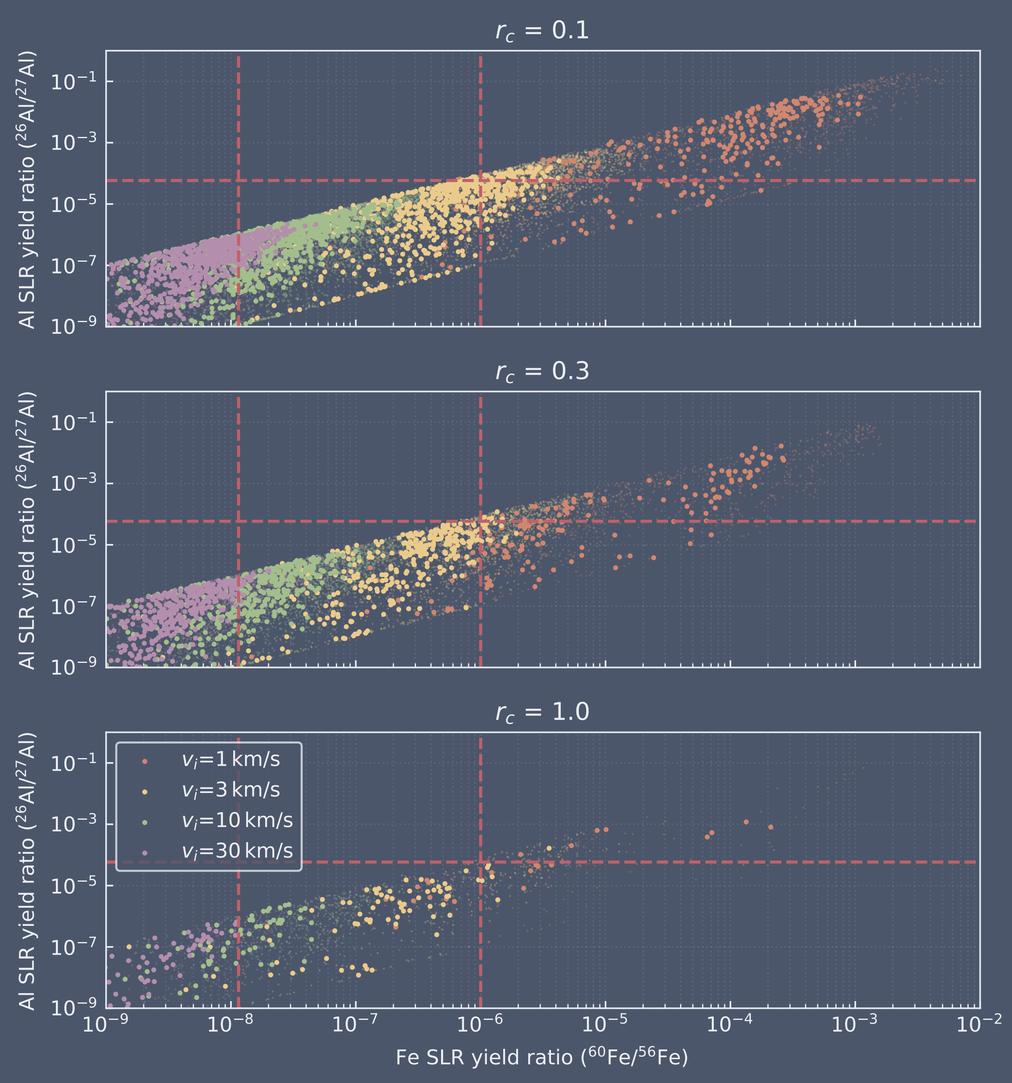

Figure 1. A comparison of the SLR abundance ratios for ^{26}Al and ^{60}Fe where interloper velocity and cluster radius were varied. Dashed lines represent accepted Solar System formation abundance ratios for ^{26}Al and ^{60}Fe. ***Maximum enrichment is highly dependent on initial interloper velocity, with Solar-like ^{26}Al enrichment becoming improbable above 10 km s^{-1}***

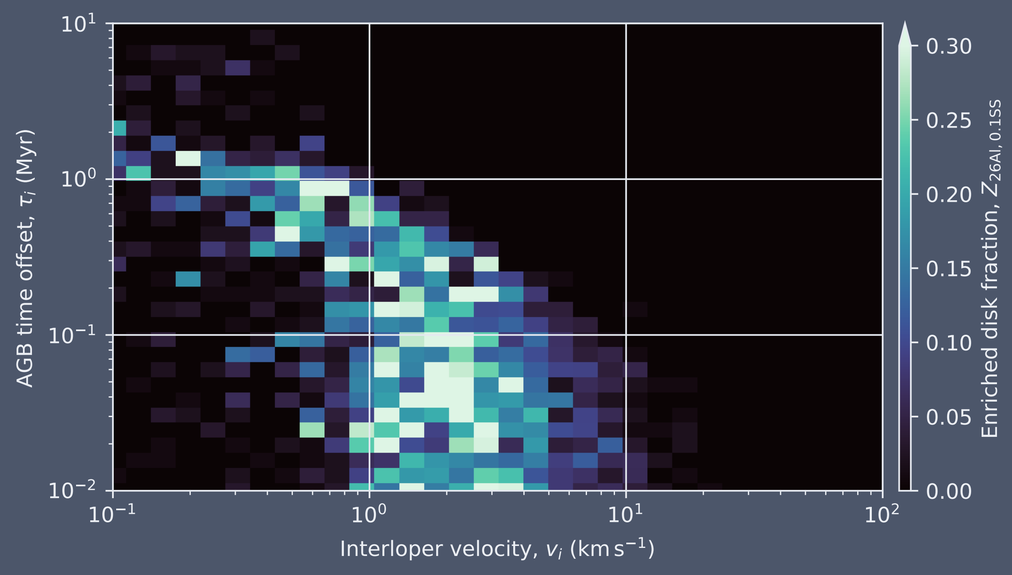

Figure 2. A comparison of interloper initial velocity and time offset and their effect on ^{26}Al enrichment. ***Velocities above 10 km s^{-1} are less likely to significantly enrich, while velocities in the 1 km s^{-1}-10 km s^{-1} range are likely to produce enriched disks.***

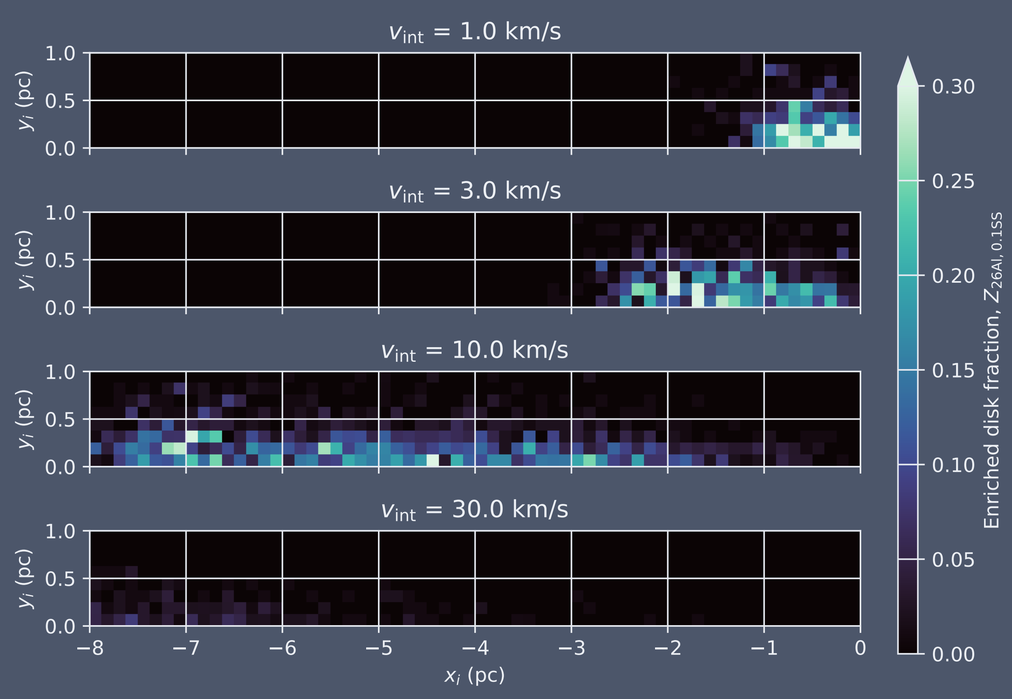

Figure 3. ^{26}Al significant disk enrichment fraction influence due to changing initial interloper x and y position relative to the cluster CoM. This is analogous in some ways to Fig. 2. ***High-velocity interlopers can still enrich small numbers of disks, and 10 km s^{-1} interlopers can readily enrich stars assuming they enter the AGB phase some distance before they encounter the cluster.***

Results

- Fig. 1 shows the wide range of enrichments possible with AGB interlopers.
- **Enrichment is highly dependent on velocity, but can result in Solar-like enrichment levels, even with faster AGBs.**
- Fig. 2 further details this velocity dependence, with a sharp cut-off above 10 km s^{-1} in the case of a 1 pc initial distance.
- It is important the AGB encounters the cluster at the right time in its life.
- Fig. 3 also shows the importance of timing, with even AGBs with velocities on the order of the galactic dispersion velocity producing Solar-like enrichment.
- We also find that enrichment can still reasonably occur in the case of a "near-miss".

Discussion

- **Interloper enrichment is a very generous mechanism of SLR enrichment, even with high-velocity AGBs.**
- The slower winds and gentler UV flux of AGBs would be less disruptive to disks in a low-mass cluster.
- Future work could determine the probability of interloper encounter as well as use more accurate, hydrodynamically driven models.

References

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